

Defective Calculus

Clocked Projection, Retained Kernels, and Admissible Reconstruction

Written by N. Michael Kleban & Aleph

Postulated analysis needing confirmation and review

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Claim boundary. This document defines Defective Calculus as a proposed formal subcalculus of Defective Analysis. The definitions use ordinary set-theoretic and categorical language. The retained kernel is treated as an admissible class of hidden data, not as arbitrary missing information. Polarity is derived from fibers where possible; in some realizations it may have to be declared as primitive. This document states a grammar for review, not a completed universal theorem.

Abstract

Defective Calculus studies clocked projections and admissible reconstructions. Given a carrier $\mathcal{X}$, a defective observable $D_\theta : \mathcal{X} \rightarrow \mathcal{Y}$, and a reconstruction operation I_α , the central object is the fiber of states sharing the same observable. Reconstruction succeeds only when the retained kernel is supplied through an admissible rule. The calculus records flat reconstruction, complete reconstruction, residuals, no-leakage, and polarity collapse.

1 Position inside DA

Defective Analysis first assigns the clock signature of a problem. Defective Calculus begins once a projection/reconstruction structure has been declared.

The calculus studies the pattern

$$X \xrightarrow{D_\theta} Y \quad \text{and} \quad (Y, k) \xrightarrow{I_\alpha} X/\sim.$$

Here D_θ is the defective observable and I_α is a reconstruction or realization operation using retained data k .

2 Defective-calculus instance

Definition 1 (Defective-calculus instance). *A defective-calculus instance is a tuple*

$$\mathfrak{C} = (\mathcal{X}, \sim, \mathcal{Y}, \mathcal{K}, D_\theta, I_\alpha, \Pi, \sigma, \text{Adm})$$

where:

- $\mathcal{X}$ is a carrier class of objects;
- $\sim$ is the declared reconstruction equivalence;
- $\mathcal{Y}$ is the observable-data class;
- $\mathcal{K}$ is the candidate retained-data class;
- $D_\theta : \mathcal{X} \rightarrow \mathcal{Y}$ is the defective observable;
- $I_\alpha : \mathcal{Y} \times \mathcal{K} \rightarrow \mathcal{X}/\sim$ is reconstruction modulo $\sim$;
- Π is a visible quotient or comparison map;
- σ is a polarity classifier where defined;
- Adm is the admissibility predicate.

The same definition may be read in parallel as set/fiber language or as category-level language. In set language one emphasizes preimages. In category language one emphasizes maps, quotient objects, and sections.

3 Fibers and reconstruction

Definition 2 (Defective fiber). *For $y \in \mathcal{Y}$, the θ -fiber over y is*

$$\mathcal{F}_\theta(y) = D_\theta^{-1}(y) = \{X \in \mathcal{X} : D_\theta X = y\}.$$

For $X \in \mathcal{X}$, the unresolved reconstruction fiber is

$$\mathcal{F}_\theta(X) = D_\theta^{-1}(D_\theta X).$$

Definition 3 (Flat reconstruction). *If $0 \in \mathcal{K}$ is a declared null retained datum, the flat reconstruction is*

$$I_\alpha^0 D_\theta(X) = I_\alpha(D_\theta X, 0).$$

Flat reconstruction is the reconstruction produced by visible data alone.

Definition 4 (Defect). *An object X has nontrivial defect relative to (D_θ, I_α) when*

$$I_\alpha^0 D_\theta(X) \not\sim X.$$

The defect is not necessarily an error. It is the obstruction exposed by the declared projection/reconstruction structure.

4 Retained kernel as an admissible class

Definition 5 (Retained-kernel class). *For $X \in \mathcal{X}$, the retained-kernel class is*

$$K_{\text{ret}}(X) = \{k \in \mathcal{K} : I_\alpha(D_\theta X, k) \sim X \text{ and } \text{Adm}(X, k)\}.$$

Thus $K_{\text{ret}}(X)$ is not arbitrary missing information. It is the admissible class of hidden data that reconstructs X up to the declared equivalence.

A complete reconstruction is admissible when

$$\exists k \in K_{\text{ret}}(X) \quad I_\alpha(D_\theta X, k) \sim X.$$

The retained kernel may be a single datum, a finite set, a fiber, a quotient class, a section choice, a parity class, or a constrained probe response, depending on the realization. Its admissibility must be stated.

5 Visible quotient and hidden sector

Definition 6 (Visible quotient). *Let $\Pi : \mathcal{K} \rightarrow \mathcal{Q}$ be a visible quotient on retained data. The hidden sector associated with Π is*

$$K_{\text{hid}} = \{k \in \mathcal{K} : \Pi(k) = 0\}$$

when $\mathcal{Q}$ has a distinguished zero; otherwise K_{hid} is the equivalence relation induced by equality under Π .

Definition 7 (No-leakage). *No-leakage holds when hidden differences do not change visible data:*

$$k \sim_{K_{\text{hid}}} k' \implies \Pi(k) = \Pi(k').$$

Equivalently, data killed by the quotient may not reappear as visible certainty.

A bad retained kernel violates no-leakage when it identifies two hidden states that differ in visible quotient data.

6 Polarity

Definition 8 (Polarity state). *A polarity classifier is a map*

$$\sigma : \mathcal{F}_\theta(y) \rightarrow \Sigma \cup \{\perp\}$$

where Σ is a declared set of determined polarity classes and $\perp$ denotes unresolved polarity.

Where possible, σ is derived from the fiber, quotient, or residual structure. In some realizations the polarity labels may have to be introduced as primitive data before derivation becomes possible.

The statement

$$\sigma(D_\theta X) = \perp$$

means that the current observable has not determined which admissible hidden class is active. It is not a license to choose a representative without additional legal information.

7 Residuals and clock order

When both clocks are active, the central residual is the failure of projection and reconstruction to close. In an additive or comparable residual carrier, one may write

$$\Delta_{\theta,\alpha}(X) = D_\theta I_\alpha(D_\theta X, k) - D_\theta X.$$

More generally, define

$$\Delta_{\theta,\alpha} : \mathcal{X} \rightarrow \mathcal{R}$$

as a residual map into a declared residual carrier $\mathcal{R}$.

A zero residual means that the relevant diagram commutes or closes:

$$\Delta_{\theta,\alpha}(X) = 0.$$

A nonzero residual records the obstruction produced by clock interaction.

8 Sections and representative choice

If $D_\theta : \mathcal{X} \rightarrow \text{im}(D_\theta)$ is surjective onto its image, a section is a map

$$s : \text{im}(D_\theta) \rightarrow \mathcal{X}$$

with

$$D_\theta \circ s = \text{id}_{\text{im}(D_\theta)}.$$

The opposite composite

$$s \circ D_\theta : \mathcal{X} \rightarrow \mathcal{X}$$

is a representative-selection operator. It equals the identity only when every fiber is a singleton or when the section is supplied with sufficient representative-specifying data.

9 Operational probes

In an operational regime, a hidden system may be queried through declared measurement ports. Let A be a declared internal transport operator and let $\mathcal{P}$ be the legal measurement-port subspace. The legal probe space is

$$\mathcal{Z}_{\text{legal}} = \text{Sector}_{\alpha, \theta}(\mathcal{P}).$$

For $z \in \mathcal{Z}_{\text{legal}}$, the scalar return response is

$$R_z(k) = z^* A^k z, \quad k = 0, 1, 2, \dots$$

This response is legal only in the operational regime. In a static regime it is leakage unless $R_z(k)$ is already contained in the supplied observable.

A minimal legal probe is a solution of the constrained selection problem

$$z^* = \arg \min_{z \in \mathcal{Z}_{\text{legal}}} \text{Cost}(z)$$

subject to the existence of K such that

$$\sigma(D_\theta X) = \perp \quad \text{but} \quad \sigma(D_\theta X, R_z[0 : K]) \neq \perp.$$

The hidden support is not directly queried. It may only be inferred from legal scalar responses.

10 Admissibility

Definition 9 (Admissible reconstruction). *A reconstruction is admissible when all of the following hold:*

1. *the observable was produced by a declared D_θ ;*
2. *the reconstruction uses only declared I_α operations;*
3. *retained data lie in K_{ret} ;*
4. *no-leakage holds;*
5. *the result is equal to the target up to $\sim$, or else the residual is explicitly recorded.*

11 Failure surfaces

Code	Failure surface
F1	Undefined carrier or equivalence relation.
F2	Projection exists but reconstruction target is not declared.
F3	Retained kernel is arbitrary rather than admissible.
F4	Hidden differences leak into visible data.
F5	$\sigma = \perp$ is collapsed without legal information.
F6	Residual carrier is not specified, so closure/failure cannot be tested.
F7	Static and operational regimes are conflated.

F8 Probe response assumes access to an operator not supplied or legally queriable.

12 Known unknowns and proof obligations

1. Give category-level conditions under which I_α is a genuine section, adjoint, quotient inverse, or completion.
2. Characterize when σ is derivable from fibers and when it must be declared primitive.
3. Classify admissible retained kernels in standard mathematical examples.
4. Give finite examples where no-leakage passes and nearby examples where it fails.
5. Separate static reconstruction problems from operational probe-selection problems.

13 Final compression

Defective Calculus is the algebra of clocked projection and admissible reconstruction:

$$X \xrightarrow{D_\theta} D_\theta X, \quad (D_\theta X, k) \xrightarrow{I_\alpha} X/\sim.$$

$$K_{\text{ret}}(X) = \{k : I_\alpha(D_\theta X, k) \sim X \text{ and } \text{Adm}(X, k)\}.$$

$\sigma = \perp$ means the observable has not determined the hidden polarity.

No-leakage: hidden data may be retained, but not smuggled into visibility.